

1. Can you confirm the proposed location of the new well (front, back, or a side of the building), specifically beyond the requirement of being within 50 feet of the existing well? Any site plans or preferred placement would be helpful.

Response: The proposed location of the new well be within 50 feet of the North-West side of the well house. Please refer to the "Site Map" Attachment

2. Is the facility currently connected to a septic system or public sewer/water system?

Response: The facility is currently connected to a septic system.

3. Are there any known underground obstructions or hazards in the work area, such as buried fuel or oil tanks?

Response: There are no buried fuel or oil tanks. However, there are a series of buried, active communications lines and electrical lines. Also, there are some old abandoned, buried concrete footers from a demolished water tower. The contractor should perform their own survey to avoid buried obstructions in the intended construction area for the well.

4. Are there any existing irrigation systems or auxiliary water connections tied to the current well that need to be considered?

Response: There are no existing irrigation systems. NRL does not believe auxiliary water connections are tied to the current well.

5. Can you provide an anticipated project timeline?

Response: Project completion is expected to be within 6 months from issuance of a contractual award.

6. Are existing well logs, historical yield data, or geotechnical/soil reports available for the current well?

Response: Geotechnical/soil reports are available for the current well. Please refer to the "Well Completion Report" Attachment.

7. SOW specifies to conduct final pumping tests after the well has been constructed and references meeting volume per day requirements for (5) five consecutive days. Is the Government requesting a 24/7 pump test at 10 GPM for five consecutive days? If so, please be advised this is not a typical requirement or standard practice and is incredibly wasteful. The site would be required to be manned 24/7 for the duration of the testing, incurring significant unnecessary costs, and the total quantity of wasted water that would be running off would be a minimum of 72,000 gallons. Please advise.

Response: Based on Code of Maryland Regulations (COMAR) 26.04.04.25D(2), Section 26 Minimum Yield and Yield Testing for Domestic Water Supply Wells”, domestic well or double well combination shall be tested to produce a minimum yield of 1 gallon per minute (GPM) for 6 hours.

Given that this water well project seeks a 10 GPM rate, the applicable testing should correspond to 10 GPM for 6 hours, which would equate to an estimated 3,600 gallons.

[https://regs.maryland.gov/us/md/exec/comar/26.04.04.25#D\(2\)](https://regs.maryland.gov/us/md/exec/comar/26.04.04.25#D(2))

8. If the above-mentioned 5-day test is determined to be an unwise requirement, please provide the actual testing requirements for this project.

Response: See response to question #7.

9. Please clarify the water quality testing requirements; COMAR requires 1 nitrate, 1 nitrite, and 2 BT tests. The MD/County Health Department typically is involved with testing and issuing the Certificate of Potability. If additional water tests are required, please provide the full panel of tests needed.

Response: Water quality testing should correspond to Code of Maryland Regulations (COMAR) requirements.

10. SOW states to fill annular space exterior of casing from top to bottom with concrete grout.

a. First, concrete grout cannot be used due to the high temperatures during cure, which can damage the well casing. Industry standard is to utilize cement/bentonite mixture of grout. Please confirm this is acceptable.

Response: The SOW Well Specifications updated to reflect cement/bentonite-mixture grout.

b. Second, why is grout being requested from bottom to top? COMAR regulation requires 5' into the first confining layer of clay. Please advise.

Response: Agreed, the grout is only required to extend into the first confining layer of clay. As stated in the Code of Maryland Regulations (COMAR): 26.04.04.19, Section 19 "Construction Standards – Grouting", Sub-section - Section E "Standards for Grouting Confined Aquifer Wells in Hydrogeologic Area 2", Items (5) and (6) are applicable:

(5) For wells intended to have a yield greater than 5,000 gallons per day, the grout shall extend from a minimum of 5 feet into the confining bed, immediately above the aquifer being used, to the land surface.

(6) For wells intended to have a yield of less than 5,000 gallons per day, the grout shall extend from a minimum of 5 feet into the first confining bed, which is at least 5 feet thick, to the land surface.

[https://regs.maryland.gov/us/md/exec/comar/26.04.04.18#B\(1\)](https://regs.maryland.gov/us/md/exec/comar/26.04.04.18#B(1))

11. During the site visit we attended in early January, the facilities manager mentioned being able to dispose of drilling fluids and spoils from hole on site. Please confirm this is acceptable.

Response: On site disposal of drilling fluids and spoils is approved by NRL.

12. SOW specifies well being able to provide 10 GPM flow within 1-1/4" piping at 20' elevation above well head ground surface at any point on the facility. Please advise if there are

any locations that are indeed 20' above the well head and be advised that 10 GPM flow can only be guaranteed if the entire run of water piping is indeed 1-1/4".

Response: There is one occupied facility with a 20' elevation (Building A-9). NRL cannot currently confirm if the entire existing run of water piping is 1-1/4".